

Target Vocabulary_Energy Sources

Furnace n. /'fɜ:nɪs/ (печь)	A device used for heating, often for industrial processes, or for providing warmth in a building.	The steel mill utilized a blast furnace to melt and refine the raw iron ore.
Electrify v. /'iːlektrɪfaɪ/ (электрифицировать)	To power or equip with electricity; to convert a system or process to operate using electricity.	The railway company decided to electrify the train lines for improved efficiency and reduced environmental impact.
Demand n. /dɪ'ma:nd/ (спрос, потребность)	The amount of electrical power required by consumers within a specific timeframe, often influenced by factors such as population, economic activity, and technological advancements.	During the heatwave, the electricity demand soared as air conditioning units ran at full capacity.
Infinite adj. /'ɪnfɪnət/ (бесконечный) ant finite adj /'faɪ.naɪt/ (конечный, ограниченный)	Without limits or bounds; endless.	The universe is often described as having infinite possibilities and mysteries. The earth's natural resources are finite , and we must use them judiciously.
Grid n. /ɡrɪd/ (электросеть)	Interconnected system of power lines, substations, and other equipment that transport electricity from power plants to consumers. <i>e.g. to join the grid / to be connected to the grid / to go off the grid</i>	The expansion of renewable energy sources requires upgrades to the grid to accommodate the variable nature of wind and solar power.
Release v. /rɪ'li:s/ (высвобождать)	To allow stored energy to be released, often in a controlled manner.	The dam's gates were opened to release energy from the stored water, generating electricity in the process.

Harness v. <i>/ˈhɑːnɪs/</i> (использовать, задействовать)	To capture, control, and utilize energy from a specific source, such as sunlight, wind, or water.	The wind farm harnessed energy from the strong coastal winds to generate clean electricity.
Generate v. <i>/ˈdʒenəreɪt/</i> (вырабатывать, генерировать)	To produce or create energy, often through the conversion of one form of energy to another.	The geothermal power plant generated energy by harnessing heat from beneath the earth's surface.
Sustain v. <i>/səˈsteɪn/</i> (поддерживать)	To maintain or support over time; to endure or uphold. <i>adj</i> sustainable (устойчивый) <i>n.</i> sustainability (устойчивость)	The conservation efforts aimed to sustain the fragile ecosystem for future generations.
Eliminate v. <i>/ɪˈlɪmɪneɪt/</i> (устранять)	To completely remove or get rid of something.	The company implemented a waste reduction program to eliminate unnecessary packaging and minimize environmental impact.
Intermittent adj. <i>/ˌɪntəˈmɪtənt/</i> (непостоянный, прерывистый)	Energy sources that are not continuously available, such as solar and wind power, which depend on weather conditions or time of day.	The intermittent nature of solar energy production requires effective energy storage solutions for consistent power supply.
(Nuclear waste) disposal <i>/dɪˈspəʊzəl/</i> (утилизация ядерных отходов)	The management and safe disposal of radioactive waste produced by nuclear power plants and other nuclear processes.	The government allocated funds for research into advanced nuclear waste disposal technologies to address long-term environmental concerns.
Triple v. <i>/ˈtrɪpl/</i> (утраивать)	To increase threefold; to become three times as much or as many.	The installation of solar panels tripled the amount of electricity generated by the household.

(-in) appropriate adj. <i>/ə'prəʊpriət/</i> (подходящий; неподходящий (inappropriate))	right for a particular purpose or occasion	The outdated machinery proved to be inappropriate for the modern production requirements.
inconsistent adj <i>/ˌɪnkən'sɪstənt/</i> (непоследовательны й, нестабильный)	Not in agreement or harmony with previous actions, statements, or beliefs.	The inconsistent performance of the new software raised concerns about its reliability and compatibility.
Cost-effective adj <i>/ˌkɒst ɪ'fektɪv/</i> (экономически выгодный) <i>/ˌʌn ɪ:kə'nɒmɪkl/</i>	Producing good results without costing a lot of money; providing value for the cost incurred.	The new energy-efficient appliances proved to be a cost-effective investment, leading to reduced electricity bills.
Ineffective adj <i>/ˌɪnɪ'fektɪv/</i> (неэффективный, безрезультатный)	Not producing the intended or desired results; not successful in achieving the desired outcome.	The ineffective marketing campaign failed to attract new customers to the product.
Inefficient adj <i>/ˌɪnɪ'fɪʃnt/</i> (неэффективный, неэкономичный)	Not achieving maximum productivity with minimum wasted effort or expense; not performing optimally.	The inefficient use of resources led to unnecessary delays in the project timeline.
Insufficient adj <i>/ˌɪnsə'fɪʃnt/</i> (недостаточный)	Not as much as needed; not meeting the requirements or demands.	The current budget allocation was insufficient to cover the costs of the proposed research project.
Ensure v. <i>/ɪn'ʃʊə(r)/</i> (обеспечивать, гарантировать)	To make certain that something will happen or be the case; to guarantee or secure.	The quality control team worked diligently to ensure that all products met the highest standards before being released to the market.
Propagate v. <i>/ˈprɒpəgeɪt/</i> (распространять)	To spread or promote an idea, belief, or practice widely.	The organization aimed to propagate awareness about sustainable energy practices through educational campaigns and community outreach.

Nuclear fission <i>/ˌnjuːkliə ˈfɪʃn/</i> (деление атомного ядра)	The process in which the nucleus of an atom splits into two or more smaller nuclei, releasing a large amount of energy.	Nuclear fission is the principle behind the energy production in conventional nuclear power plants.
Nuclear fusion <i>/ˌnjuːkliə ˈfjuːʒn/</i> (ядерный синтез)	The process in which two atomic nuclei combine to form a heavier nucleus, releasing a tremendous amount of energy.	Nuclear fusion is the process that powers the sun and other stars, producing vast amounts of energy through the fusion of hydrogen atoms.

Kinetic adj <i>/kɪˈnetɪk/</i> (кинетический)	of or produced by movement	kinetic energy
Potential adj <i>/pəˈtenʃl/</i> (потенциальный)	that can develop into something or be developed in the future	potential energy
Atomic adj <i>/əˈtɒmɪk/</i> (атомный)	- relating to the energy that is produced when atoms are split; related to weapons that use this energy - relating to atoms or an atom	atomic energy/power atomic structure
Thermal adj <i>/ˈθɜːml/</i> (тепловой)	connected with heat	thermal energy
Hydroelectric adj <i>/ˌhaɪdrəˈɪlektrɪk/</i> (гидроэлектрический)	using the power of water to produce electricity; produced by the power of water	a hydroelectric plant

Solar <i>adj</i> /'sɒlə(r)/ (солнечный)	1. using the sun's energy 2. of or connected with the sun	1. a satellite telephone with solar -powered batteries 2. solar radiation
Tidal <i>n.</i> /'taɪdl/ (приливный)	connected with tides (= the regular rise and fall of the sea)	tidal forces/energy
Replenish <i>v.</i> /rɪ'plenɪʃ/ (пополнять)	to make something full again by replacing what has been used	to replenish food and water supplies
Retain <i>v.</i> /rɪ'teɪn/ (сохранять, удерживать)	to keep something; to continue to have something to continue to hold or contain something	These plants will need a soil that retains moisture during the summer months.
Evaporate <i>v.</i> /ɪ'væp.ər.eɪt/ (испарять, испаряться)	to cause a liquid to change to a gas, especially by heating	The high concentration of sugars forms a syrup when the sap evaporates .
Fuse <i>v.</i> /fju:z/ (соединять, сплавлять; перегорать)	1. to join together physically, or to join things together physically 2. to melt or make two things melt together, especially at a high temperature 3. When an electrical device or piece of machinery fuses, or when someone or something fuses it, it stops working because the electric current is too high	1. The bones of the skull are not properly fused at birth. 2. The heat of the fire fused many of the machine's parts together 3. Either my headlights have fused or the bulbs have gone.

Odorous <i>adj.</i> /'əʊdərəs/ (пахучий, имеющий запах) <i>Ant. odorless</i> (без запаха)	having a smell	odorous gases
Conventional <i>adj.</i> /kən'venʃənl/ (традиционный, обычный)	following what is traditional or the way something has been done for a long time	You can use a microwave or cook it in a conventional oven.
Decarbonize <i>v.</i> /di:'kɑ:bənaɪz/ (декарбонизировать)	to replace fossil fuels with a fuel that is less harmful to the environment	If we decarbonize electricity then through electric cars we can decarbonize transport.
Friction <i>n.</i> /'frɪkʃn/ (трение)	the resistance (= the force that stops something moving) of one surface to another surface or substance moving over or through it	The force of friction slows the spacecraft down as it re-enters the earth's atmosphere.
Torsion <i>n.</i> /'tɔ:ʃn/ (кручение)	twisting of a part by applying equal and opposite torques at either end.	Ways to put an item into torsion include holding one end of it still while you twist the other end, or twisting both ends in opposite directions.
Compression <i>n.</i> /kəm'preʃn/ (сжатие)	the action of applying a force to a solid, liquid, or gas so that it takes up less space.	A column will buckle under compression if it is too slender.
Expansion <i>n.</i> /ɪk'spænʃn/ (расширение)	an act of increasing or making something increase in size, amount or importance	Thermal expansion is the tendency of matter to change its shape, area, volume, and density in response to a change in temperature.
Centrifugal force /,sentri'fju:gl/ (центробежная сила)	the force that makes objects move outwards when they are spinning around something or travelling in a curve.	The juice is extracted by centrifugal force .

Tension n. <i>/ˈtɛnʃn/</i> (натяжение, напряжение)	the state of being stretched tight; the extent to which something is stretched tight	muscular tension
Electron n. <i>/ɪˈlektɹɒn/</i> (электрон)	a very small piece of matter (= a substance) with a negative electric charge, found in all atoms	Electric current is the flow of electrons in an electrical conductor.
Generator n. <i>/ˈdʒenəreɪtə(r)/</i> (генератор)	a machine for producing electricity	The factory's emergency generators were used during the power cut.
Spin v. <i>/spɪn/</i> (вращать, вращаться)	to turn round and round quickly; to make something do this	to spin a ball/coin/wheel
Cable n. <i>/ˈkeɪbl/</i> (кабель)	a set of wires, covered in plastic or rubber, that carries electricity, phone signals, etc.	fibre-optic cable
Joule n. <i>/dʒuːl/</i> (джоуль)	a unit of energy or work	Cellphones can have a maximum power output of 2W, which is 2 joules per second.
Power v. <i>/ˈpaʊə(r)/</i> (снабжать энергией)	to supply a machine or vehicle with the energy that makes it work	The plant generates enough energy to power a town of 6 000 people.
Pump n. <i>/pʌmp/</i> (насос)	a machine that is used to force liquid, gas or air into or out of something	a gas pump